Math-M-Addicts
MINDS THAT COUNT

SP250 Homework 3

1. Let a, b, c, d, m with $m \neq 0$ be integers. Suppose that $a \equiv b \pmod{m}$.

   a) Prove that $ac \equiv bc \pmod{m}$.

   b) Now suppose additionally that $c \equiv d \pmod{m}$. Prove that $ac \equiv bd \pmod{m}$.

Solution:

a) $a \equiv b \pmod{m}$ means that $m \mid (a - b)$ such that $a - b = mk$ for some integer k. Now $ac - bc = (a - b)c = mkc$, so $m \mid (ac - bc)$. Therefore, $ac \equiv bc \pmod{m}$.

b) Applying the result of a), we have $ac \equiv bc \pmod{m}$ as well as $bc \equiv bd \pmod{m}$. Hence, $ac \equiv bc \equiv bd \pmod{m}$.

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2. a) Prove that out of any 10 integers, one can pick two, whose difference is a multiple of 9.

b) Fanny picked several integers such that no two of them add up to a multiple of 9 and no two differ by a multiple of 9. What is the maximum number of integers that Fanny could have picked? Explain your answer.

Solution:

a) When an integer is divided by 9, there are 9 possible remainders: 0, 1, 2, 3, 4, 5, 6, 7, 8. By the pigeonhole principle, among any 10 integers, there must be two that have the same remainder when divided by 9. The difference between such two integers is a multiple of 9.

b) We separate all integers into 5 groups based on their remainders when divided by 9:
Group 1: remainder 0
Group 2: remainder 1 or 8
Group 3: remainder 2 or 7
Group 4: remainder 3 or 6
Group 5: remainder 4 or 5
Two integers in the same group will either add up to a multiple of 9 or differ by a multiple of 9. This means that Fanny can pick at most one integer from each group, or at most 5 integers total. One way to pick 5 such integers is {5, 6, 7, 8, 9}. Hence, the maximum number of integers that Fanny can pick is 5.

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3. 35 people have gathered for a party. Some of them shook hands. Each person wrote down on a piece of paper the number of hands they shook. Could the sum of all the numbers be equal to 239? Prove your answer.

Solution:

Each handshake is counted by both of the people involved, so the sum of all the numbers that people wrote

down must be two times the number of handshakes. Therefore, the sum must be even and cannot be 239.

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4. 25 Students live in a dorm, which is shaped as a 5x5 square. One day, simultaneously, some of them decide to switch rooms in such a way that every student would move to a room that is neighboring his or her own (neighboring rooms are rooms that **share a side**). Prove that at least one student will end up staying in their own room.

Solution:
Let's assign the colors black and white to the dorm rooms in the chess board pattern shown below. Under this coloring scheme, every student that moves will switch to a room with a different color. Since there are 13 black rooms and 12 white rooms, it is not possible for all the students in the black rooms to switch to a white room. Hence, at least one student will have to stay in their own room.



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5. Does there exist a 10-digit number, consisting only of digits 2,5 and 8, that is a perfect square?

Solution:
No. Recall that any number mod 3 is the same as the sum of its digits mod 3. Since 2, 5, and 8 are all 2 mod 3, any 10-digit number consisting only of digits 2, 5, and 8 will be 20 mod 3, which is equivalent to 2 mod 3. However, perfect squares can only be 0 or 1 mod 3. Hence, such a number cannot be a perfect square.

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6. All 10 digits from 0 to 9 are written on the board in some order. Sergey wants to put +'s and -'s (9 total, one between each consecutive pair of digits) such that the result evaluates to 0. Show that Sergey will

never be able to achieve his task, no matter what order the digits are written in and no matter what signs he uses.

Solution:
Let's consider how parity (whether a number is even or odd) changes when numbers are added or subtracted. Starting with some existing value, the result of adding or subtracting an odd number will have the opposite parity as that of the existing value, while the result of adding or subtracting an even number will have the same parity as that of the existing value. As there are an odd number of odd digits, the final result will always be odd. Since 0 is even, it can never be the result no matter what order the digits are written in and what signs are used.
